IBM APPLIED DATA SCIENCE CAPSTONE

Falcon 9 First-Stage Landing Prediction

Data collection, analysis, model results, and interactive evidence

EVIDENCE STATUS: Scrape coverage complete through the published October 2026 history; live API data unavailable; genuine Dash screenshots pending.

Prepared from project execution outputs | 30 October 2026

Executive summary

The capstone preserves a 90-launch modeling dataset and separately completes a live Wikipedia collection of 699 unique Falcon 9 flights (flight numbers 1–699) across 2010–2026. The model dataset has a 66.7% landing-success rate. A stratified 23-record test set gave Decision Tree 86.96% accuracy and 0.903 F1; Logistic Regression scored 82.61% accuracy and 0.882 F1. Seven SQLite queries, EDA figures, a Folium map capture, and verified Dash callbacks/routes are included.

Limits are explicit: 57 of the 699 scraped launches have unknown payload mass; the SpaceX API returned no JSON on the latest attempt (HTTP 522); the 90-row model sample is small and imbalanced; and genuine Dash browser screenshots could not be captured because Chrome timed out connecting to localhost.

1. Introduction and objectives

The objective is to examine the supplied Falcon 9 first-stage landing dataset and evaluate baseline classifiers for `Class` (0 = unsuccessful, 1 = successful). Work includes data quality checks, outcome analysis by site/payload/orbit/flight/reuse, SQL summaries, interactive mapping, model comparison, and a Plotly Dash application.

Repository and methodology sources

GitHub repository: https://github.com/bridrane168/SpaceX_capstone

SpaceX_Capstone_Basics_Final.ipynb

web/collect_spacex_api.py

web/scrape_launches.py

web/eda_analysis.py

web/sql_analysis.py

web/spacex_launch_sites.html

web/train_models.py

dashboard/dash_app.py

2. Data collection and data-wrangling methodology

SpaceX API collection

The collector targets https://api.spacexdata.com/v4/launches using bounded requests and JSON normalization. Its preserved initial run made 2 attempts; DNS resolution failed before an HTTP response, so the saved status is null, `api_output/launches_raw.json` is 0 bytes, and `api_output/launches.csv` is absent. A later network-enabled GET returned HTTP 522 with an HTML error body (7259 bytes), so JSON parsing and data export were not performed. The exact response is `api_output/live_retry_response.txt`; the initial DNS error remains in `api_output/collection_status.json` and `api_output/launches_error.txt`.

Wikipedia web scraping

The scraper sends a student-project User-Agent and parses the main current page plus four linked archive pages. All five live requests returned HTTP 200. It inspected 113 HTML tables and selected 36 tables matching the launch-history schema. The page family covers published past Falcon 9 launches from 2010 through the current 2026 section; year 2011 has no listed launch. The data are Falcon 9 only: nonnumeric Falcon Heavy identifiers are excluded.

Collection check	Verified result
Pages / HTTP status	5 pages; HTTP 200 for each
Inspected / selected tables	113 / 36
Launch rows / unique keys	699 / 699
Flight number range	1–699
Duplicate keys / rows	0 / 0
Missing payload mass	57
Excel workbook	113 sheets; refreshed with 113 inspected tables across 5 source pages using openpyxl

Records by launch year

Year	Launches	Year	Launches
2010	2	2012	2
2013	3	2014	6
2015	7	2016	8
2017	18	2018	20
2019	11	2020	26
2021	31	2022	60
2023	91	2024	132
2025	165	2026	117

Normalization removes reference markers, parses UTC launch timestamps, attaches continuation descriptions to their launch rows, aggregates multiple payload rows per flight/time, derives numeric payload mass where reported, and removes duplicate flight/time keys. `source_year` and `source_table_index` are integers, `flight_number` is string, `payload_mass_kg` is float, and descriptors are text. The 57 null payload masses remain missing; no value is imputed in the scraped-history export. The earlier 285-row 2025–2026 partial extract is retained as `falcon_launch_history_partial_2025_2026.csv` for traceability.

3. Data quality and modeling preparation

The model and dashboard use the preserved `spacex_cleaned_data.csv`, not the 699-row scrape. This keeps the prediction analysis aligned with the provided capstone dataset. It contains 90 rows × 18 columns: 30 class-0 and 60 class-1 records, no exact duplicate rows, and 26 missing values confined to `LandingPad`. Rows were retained. Numeric predictors use median imputation and standard scaling; categorical predictors use most-frequent imputation and one-hot encoding, fitted inside each training pipeline.

4. EDA and data-visualization methodology

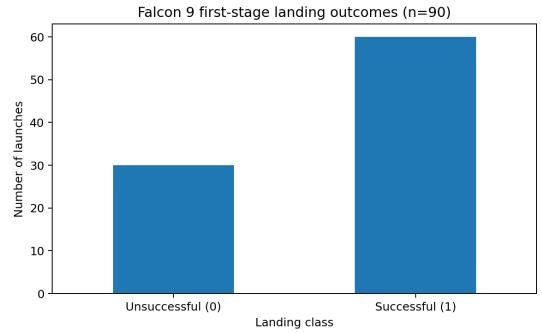
Executed summaries compare binary outcomes and success rates across launch site, payload mass, orbit, flight-number bands, and whether the booster was previously flown. Counts accompany rates so small groups remain visible. Histograms and bar charts describe the observed 90-row sample; associations are descriptive, not causal.

Launch sites

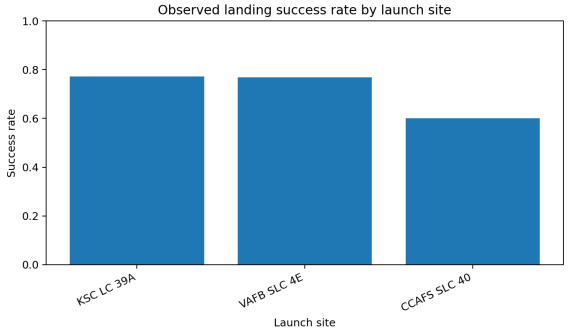
Site	Launches	Successful	Observed success
CCAFS SLC 40	55	33	60.0%
KSC LC 39A	22	17	77.3%
VAFB SLC 4E	13	10	76.9%

Payload mass by observed outcome

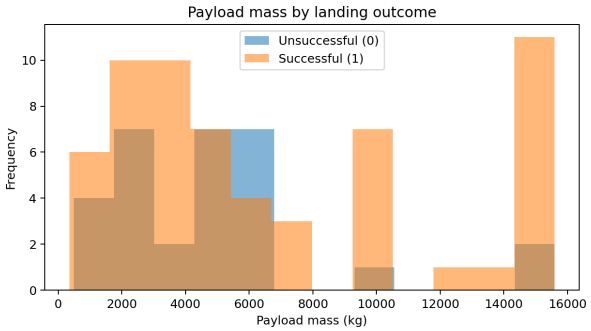
Class	Records with mass	Mean kg	Minimum kg	Maximum kg
0	30	4784.56	500.0	15600.0
1	60	6765.16	350.0	15600.0



Outcome distribution: 60 successful and 30 unsuccessful launches.



Observed landing-success rate by site; KSC 17/22, VAFB 10/13, CCAFS 33/55.

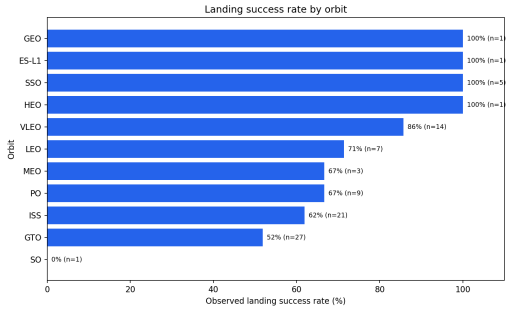


Payload mass distributions by outcome; the group means are descriptive.

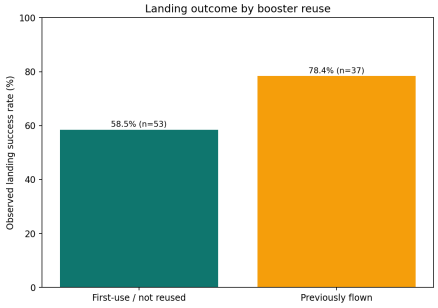
Orbit, flight-number, and booster-reuse findings

Orbit	Launches	Successes	Rate	Mean payload kg
GTO	27	14	51.9%	5,012
ISS	21	13	61.9%	3,280
VLEO	14	12	85.7%	15,316
PO	9	6	66.7%	7,584
LEO	7	5	71.4%	3,883
SSO	5	5	100.0%	2,060
MEO	3	2	66.7%	3,987
ES-L1	1	1	100.0%	570
GEO	1	1	100.0%	6,105
HEO	1	1	100.0%	350
SO	1	0	0.0%	6,105

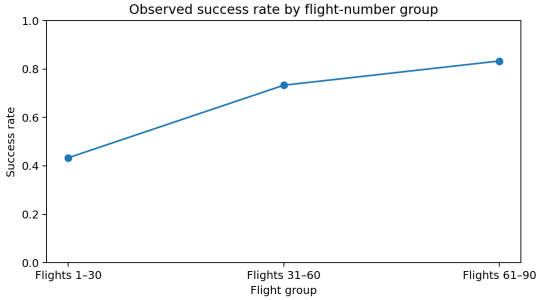
For orbit groups with at least five records, GTO has 27 launches (14 successes; 51.9%), ISS 21 (13; 61.9%), VLEO 14 (12; 85.7%), PO 9 (6; 66.7%), LEO 7 (5; 71.4%), and SSO 5 (5; 100%). The 100% rate in SSO is based on only five records. Previously flown boosters: 29/37 successes (78.4%); first-use/not-reused: 31/53 (58.5%). Flight groups 1–30, 31–60, 61–90: 13/30 (43.3%), 22/30 (73.3%), 25/30 (83.3%). Changing mission mix and small samples prevent causal claims.



Observed landing-success rate by orbit; count labels indicate group size.



Observed outcomes by booster reuse; sample counts are shown.



Observed success by flight-number group.

5. EDA with SQL analysis

The SQL runner creates an in-memory SQLite database from the cleaned 90-row CSV, loads the table, executes seven queries, and writes paired `.sql` and `.csv` outputs under `sql_output/`. These are executed results, not query text alone.

Key SQL results

Query	Verified result
Total launches	90
Binary outcomes	Class 0: 30 (33.3%); Class 1: 60 (66.7%)
Launches by site	CCAFS: 55; KSC: 22; VAFB: 13
Mean payload by outcome	Class 0: 4,784.56 kg; Class 1: 6,765.16 kg
Landing outcome × booster recovery	8 groups; see launch_outcomes.csv
Orbit-level launches, success, and mean mass	11 groups; see orbit_analysis.csv

Success rates by site

Launch site	Total launches	Success rate
KSC LC 39A	22	77.3%
VAFB SLC 4E	13	76.9%
CCAFS SLC 40	55	60.0%

Payload analysis

Class	Records with payload	Mean kg	Min kg	Max kg
0	30	4784.56	500.0	15600.0
1	60	6765.16	350.0	15600.0

Landing outcome by recovery result

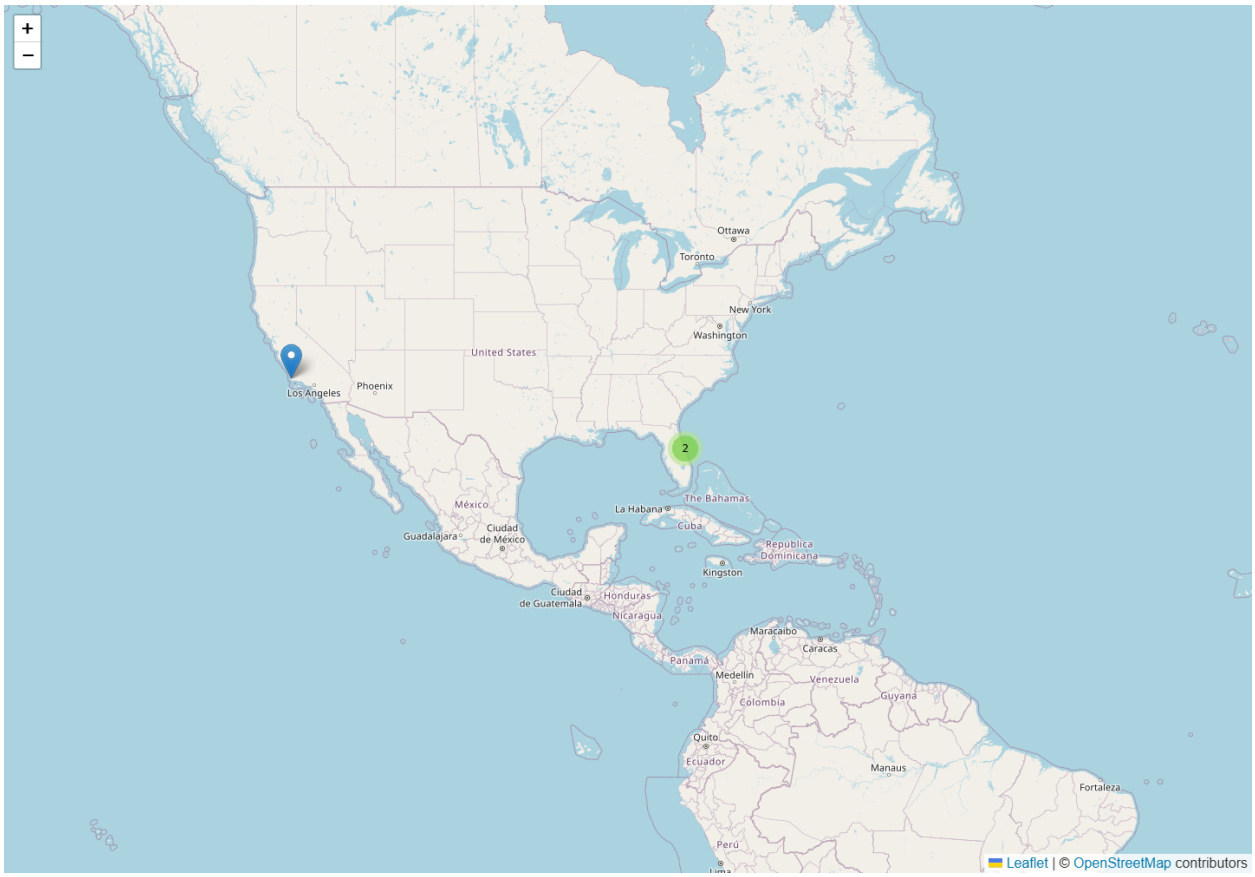
Outcome	Class	Launches
False ASDS	0	6
False Ocean	0	2
False RTLS	0	1
None ASDS	0	2
None None	0	19
True ASDS	1	41
True Ocean	1	5
True RTLS	1	14

SQL orbit output

Orbit	Launches	Success rate	Mean payload kg
GTO	27	51.9%	5,011.99
ISS	21	61.9%	3,279.94
VLEO	14	85.7%	15,315.71
PO	9	66.7%	7,583.67
LEO	7	71.4%	3,882.84
SSO	5	100.0%	2,060.00
MEO	3	66.7%	3,987.00
SO	1	0.0%	6,104.96
HEO	1	100.0%	350.00
GEO	1	100.0%	6,104.96
ES-L1	1	100.0%	570.00

6. Folium interactive map

The Folium HTML uses the project launch-site coordinates, clustered markers, popup counts, and observed site success rates. Leaflet 1.9.3, jQuery 3.7.1, and MarkerCluster 1.1.0 are bundled locally; OpenStreetMap basemap tiles still require internet. Browser verification observed Leaflet loaded, two visible marker icons, one cluster containing two Florida sites, 30 loaded map tiles, and no script errors.



Genuine browser capture of the interactive Folium map; California marker and Florida two-site cluster are visible.

7. Predictive analysis: classification methodology

The target is `Class`; selected inputs are FlightNumber, PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount, Orbit, LaunchSite, and Serial. A stratified 75/25 split produced 67 training and 23 test observations (random state 42). Numeric and categorical preprocessing is fit within each model pipeline on the training subset only. Logistic Regression and a depth-5 Decision Tree use class balancing. Held-out predictions supply the metrics; the test set contains 15 successful and 8 unsuccessful cases. Possible leakage remains because some features, such as landing configuration, may not be known before launch.

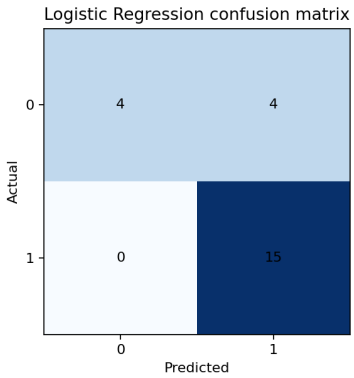
Verified predictive results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	TN FP FN TP
Logistic Regression	82.61%	0.789	1.000	0.882	0.850	4 4 0 15
Decision Tree	86.96%	0.875	0.933	0.903	0.842	6 2 1 14

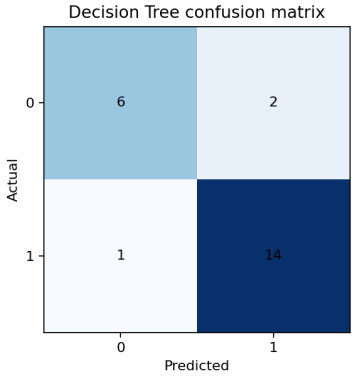
Decision Tree was selected by held-out F1 (0.903 vs. 0.882). Logistic Regression had higher ROC-AUC (0.850 vs. 0.842). Both estimates are uncertain because the test set has only 23 records.

Per-class, macro, and weighted classification results

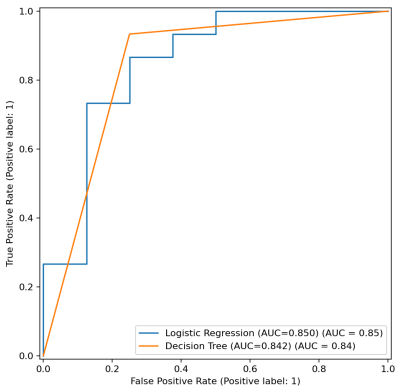
Model	Class / average	Precision	Recall	F1	Support
Logistic Regression	Unsuccessful	1.000	0.500	0.667	8
Logistic Regression	Successful	0.789	1.000	0.882	15
Logistic Regression	macro avg	0.895	0.750	0.775	23
Logistic Regression	weighted avg	0.863	0.826	0.807	23
Decision Tree	Unsuccessful	0.857	0.750	0.800	8
Decision Tree	Successful	0.875	0.933	0.903	15
Decision Tree	macro avg	0.866	0.842	0.852	23
Decision Tree	weighted avg	0.869	0.870	0.867	23



Logistic Regression confusion matrix (TN=4, FP=4, FN=0, TP=15).



Decision Tree confusion matrix (TN=6, FP=2, FN=1, TP=14).



ROC curves computed from held-out predictions.

8. Plotly Dash interactive dashboard

`dashboard/dash_app.py` is the Plotly Dash application, separate from the standalone HTML fallback. It loads the preserved 90-row CSV from a path relative to the app source. The layout provides four KPI cards (total launches, successful landings, unsuccessful landings, success rate), site and orbit filters, and five charts (outcome, site success, payload, orbit, and flight number).

Verified callback selection	Total / success / unsuccessful / rate	Chart behavior
All	90 / 60 / 30 / 66.7%	5 charts
KSC LC 39A	22 / 17 / 5 / 77.3%	5 charts
GTO	27 / 14 / 13 / 51.9%	5 charts
KSC LC 39A + GEO (empty)	0 / 0 / 0 / 0.0%	5 explicit empty-state charts

Dash 4.5.0 was installed; `/`, `/_dash-layout`, and `/_dash-dependencies` returned HTTP 200 through Flask's test client. KPI and chart callbacks were directly exercised. Chrome was pointed at the running local app but returned `ERR_CONNECTION_TIMED_OUT` for `127.0.0.1:8050`; therefore, no genuine Dash browser screenshot exists. The static matplotlib preview is not used as runtime evidence.

PENDING genuine Dash browser screenshots: dash_overview.png, dash_filter_launch_site.png, dash_filter_orbit.png. Capture them from the running app on a machine/browser that can reach localhost.

9. Findings, limitations, and conclusion

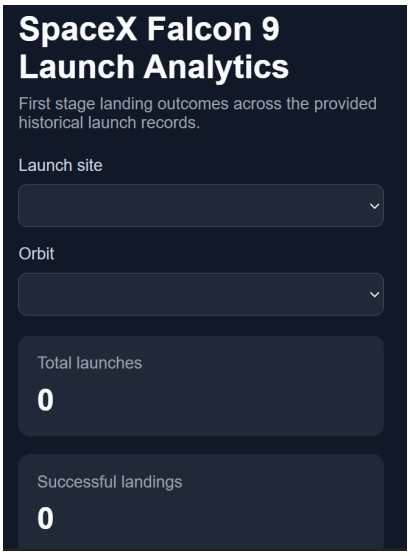
The supplied 90-row model sample shows 60 successes and 30 unsuccessful outcomes. The Decision Tree had the highest held-out F1, while Logistic Regression had slightly higher ROC-AUC. Rates vary by site, orbit, flight group, and booster reuse; these patterns are descriptive and confounded by changing missions and sample size. The larger 699-flight scrape improves historical coverage, but it is a separate web-sourced dataset and was not substituted into the course model.

Data-collection scope is improved: live Wikipedia data were fetched from five pages (HTTP 200 each) and combined into 699 distinct Falcon 9 flights spanning 2010–2026. Fifty-seven payload masses remain unknown. The SpaceX API is not represented as successful: earlier requests failed DNS and a later live request got HTTP 522 without JSON. Dash callbacks and routes work, while genuine browser screenshots remain pending due localhost transport failure.

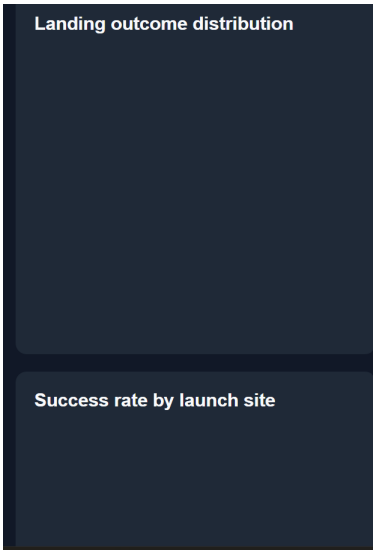
Status	Component	Evidence / remaining action
COMPLETED AND VERIFIED	Full published Wikipedia scrape	699 unique records; five pages; 113 tables; HTTP 200
COMPLETED AND VERIFIED	EDA, SQL, model results	Executed outputs and verified predictions in project
COMPLETED AND VERIFIED	Folium map screenshot	screenshots/folium_map.png; rendered tiles and cluster
COMPLETED AND VERIFIED	Dash source, callbacks, KPIs, routes	Dash 4.5.0 and outputs/dashboard_validation.txt
PENDING	Three genuine Dash screenshots	Capture in a browser with localhost access
PENDING	SpaceX API JSON/CSV	Latest HTTP 522 returned HTML; retry when endpoint is reachable
FAILED IN THIS ENVIRONMENT	Chrome to local Dash host	ERR_CONNECTION_TIMED_OUT at 127.0.0.1:8050

Appendix A. Screenshots supplied for review

The following images were supplied by the project owner and are preserved in the screenshots folder. They document notebook outputs and the local HTML preview. The HTML preview images show zero KPI values and blank plots; they are not represented as successful Plotly Dash runtime evidence. The actual Dash callback results and browser connection limitation are documented in Section 8.



Owner-supplied dashboard preview: KPI cards show zero; this is not live Dash evidence.



Owner-supplied dashboard preview: chart panels are blank; this is not live Dash evidence.

```python
from pathlib import Path
import json
status_path = Path("api_output/collection_status.json")
if status_path.exists():
    api_status = json.loads(status_path.read_text(encoding="utf-8"))
    raw_path = Path("api_output/launches_raw.json")
    csv_path = Path("api_output/launches.csv")
    print("Original collector status (two attempts):", json.dumps(api_status, indent=2))
    print("Saved raw response bytes:", raw_path.stat().st_size if raw_path.exists() else "absent")
    print("Launch CSV:", "present" if csv_path.exists() else "absent")
retry_path = Path("outputs/external_retry_evidence.json")
if retry_path.exists():
    print("Most recent genuine external retry:", json.dumps(json.loads(retry_path.read_text(encoding="utf-8")), indent=2))
```

```
Original collector status (two attempts): {
  "url": "https://api.spacexdata.com/v4/launches",
  "timestamp_utc": "2026-10-30T08:19:31.161587+00:00",
  "http_status": null,
  "attempts": 2,
  "success": false,
  "error": "ConnectionError: HTTPSConnectionPool(host='api.spacexdata.com', port=443): Max retries exceeded with url: /v4/launches (Caused by NameResolutionError(\"<urllib3.connecti
}
Saved raw response bytes: 0
Launch CSV: absent
Most recent genuine external retry: [
  {
    "source": "wikipedia",
    "url": "https://en.wikipedia.org/wiki/list_of_Falcon_9_and_Falcon_Heavy_launches",
    "timestamp_utc": "2026-10-30T09:55:29.691312+00:00",
    "http_status": 200,
    "reason": "OK",
    "content_type": "text/html; charset=UTF-8",
    "response_bytes": 3153476,
```

Owner-supplied notebook capture showing preserved DNS failure and later HTTP retry evidence.

```
 Logistic Regression
Accuracy: 0.8260869565217391
Precision: 0.7894736842105263
Recall: 1.0
F1: 0.8823529411764706
Confusion matrix [actual rows 0,1; predicted columns 0,1]:
 [[ 4  4]
 [ 0 15]]
              precision    recall  f1-score   support

           0       1.00      0.50      0.67         8
           1       0.79      1.00      0.88        15

    accuracy                           0.83        23
   macro avg       0.89      0.75      0.77        23
weighted avg       0.86      0.83      0.81        23

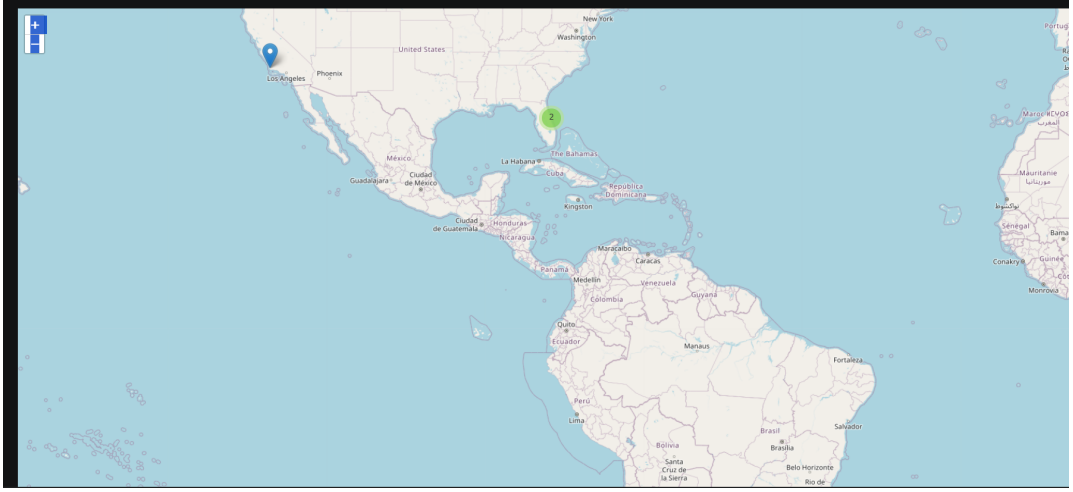

 Decision Tree
Accuracy: 0.8695652173913043
Precision: 0.875
Recall: 0.9333333333333333
F1: 0.9032258064516129
Confusion matrix [actual rows 0,1; predicted columns 0,1]:
...
    accuracy                           0.87        23
   macro avg       0.87      0.84      0.85        23
weighted avg       0.87      0.87      0.87        23

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Owner-supplied notebook capture of classification metrics and test-set output.

```
display(m)
```

```
Interactive map generated successfully!
Total launch sites: 3
```



Owner-supplied notebook capture of the Folium map output; the report also includes the separately verified map screenshot.